

Volatility Surface Modelling and Forecasting

From Lévy Processes to Deep Learning

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The Data: GLD Options on 18 July 2025

A real option chain with 11 maturities and 877 quotes

Why GLD?

- Liquid gold ETF, tight bid-ask spreads
- 11 maturities: 8 DTE to 127 DTE
- ~ 80 strikes per maturity (call + put)
- Source: Barchart.com end-of-day quotes

Key parameters:

- Spot: $S_0 = \$308.39$
- Moneyness range: $K/S_0 \in [0.924, 1.064]$
- Risk-free rate: interpolated Treasury curve (4.27% – 4.46%)
- Dividend yield: $q = 0.4\%$ (GLD expense ratio)

What We Observe

The market does not trade at a single implied volatility. Instead, every strike–maturity pair carries its own $\sigma_{\text{imp}}(K, T)$. Our job is to model this **surface**.

877 contracts
11 expirations
1 common benchmark

How Do We Compute Market-Observed Implied Volatilities?

Step 1: the Black-Scholes formula.

The price of a European call under GBM is

$$C(K, T) = S_0 e^{-qT} \Phi(d_1) - K e^{-rT} \Phi(d_2)$$

and for a put, by put-call parity,

$$P(K, T) = K e^{-rT} \Phi(-d_2) - S_0 e^{-qT} \Phi(-d_1)$$

where

$$d_1 = \frac{\log(S_0/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T},$$

and $\Phi(\cdot)$ is the standard normal CDF.

Key point. This formula gives a *one-way* map:

$$\sigma \longrightarrow \text{price.}$$

The market gives us the reverse: it quotes a *price*. We must *invert* the formula to recover the volatility implied by that price.

Step 2: Inverting Black-Scholes (Newton–Raphson)

For each market quote $(S_0, K, T, r, q, P^{\text{mkt}})$ we solve

$$C^{\text{BS}}(\sigma_{\text{impl}}; S_0, K, T, r, q) - P^{\text{mkt}} = 0$$

via the Newton–Raphson iteration:

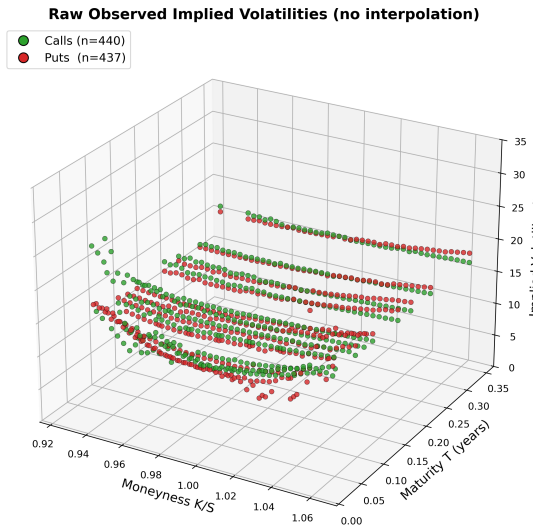
$$\sigma_{n+1} = \sigma_n - \frac{C^{\text{BS}}(\sigma_n) - P^{\text{mkt}}}{\partial C^{\text{BS}} / \partial \sigma |_{\sigma_n}}$$

where the derivative is the **Vega**:

$$\mathcal{V}(\sigma) = S_0 e^{-qT} \sqrt{\frac{T}{2\pi}} \exp\left(-\frac{1}{2}d_1(\sigma)^2\right) > 0.$$

Fallback. If Vega collapses (deep OTM, tiny T), we switch to bisection search on $[\sigma_{\min}, \sigma_{\max}] = [10^{-6}, 300\%]$.

Result: Raw Observed Implied Volatilities



What you see. Each dot is a single GLD option quote, converted from market price to implied volatility via BS inversion.

Green = calls, red = puts. 877 quotes total, moneyness $K/S \in [0.924, 1.064]$, maturities 8–127 DTE.

No model has been fitted. These are *pure market data* expressed in volatility units associated with a non-flatness property — IVs ranging from $\sim 11\%$ to $\sim 29\%$.

The problem. The quotes are irregularly spaced: dense near ATM, sparse in the wings, missing entirely in some strike–maturity regions. To reason about the surface as a whole, we need a continuous representation.

Step 3: Linear Interpolation for Visualisation

The raw data is **discrete and irregular**. To produce a continuous surface for visual inspection, we interpolate onto a regular (K, T) grid.

Method: barycentric linear interpolation (`scipy.griddata`).

For each grid point (K_i, T_j) , identify the triangle of three nearest observed quotes in the (K, T) plane and compute the barycentric weighted average of their IVs:

$$\hat{\sigma}(K_i, T_j) = \sum_{m=1}^3 w_m \sigma_m^{\text{mkt}}, \quad \sum_{m=1}^3 w_m = 1, \quad w_m \geq 0.$$

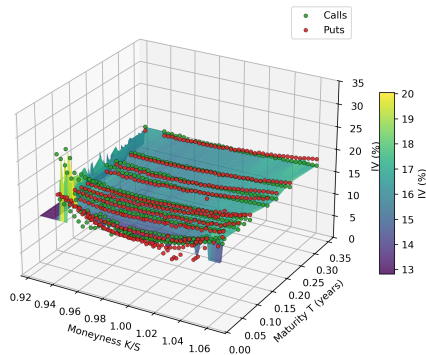
What this does NOT do:

- It is **not** a model fit — no parameters are estimated.
- It is **not** an extrapolation — grid cells outside the convex hull of observed quotes are filled with the nearest observed value (nearest-neighbour fallback).
- It makes **no assumption** about the stochastic process driving the underlying.

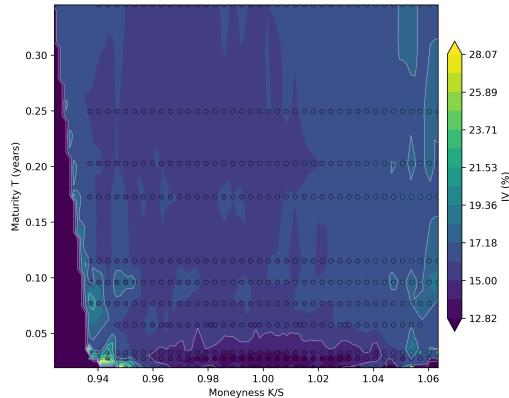
It simply connects the dots so we can see the shape of the smile.

The Baseline: Unified Observed Surface

Unified Surface: Calls + Puts Interpolated Together



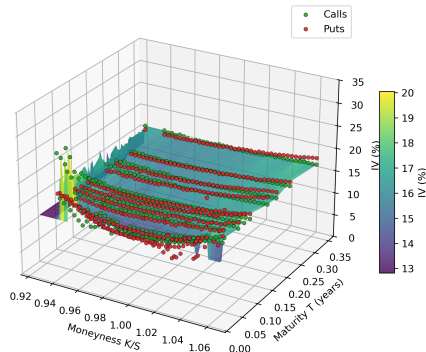
2D Contour: Calls + Puts Unified



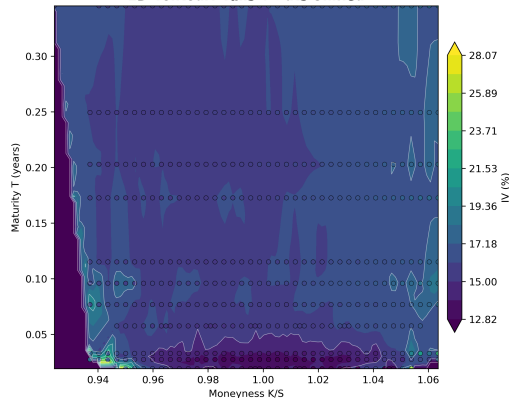
To produce a complete surface we first convert *all* puts to call-equivalent prices via put-call parity, invert Black-Scholes on the unified panel, then interpolate. The result is a single surface built from **all 877 quotes** — not calls only.

The Baseline: Unified Observed Surface

Unified Surface: Calls + Puts Interpolated Together



2D Contour: Calls + Puts Unified

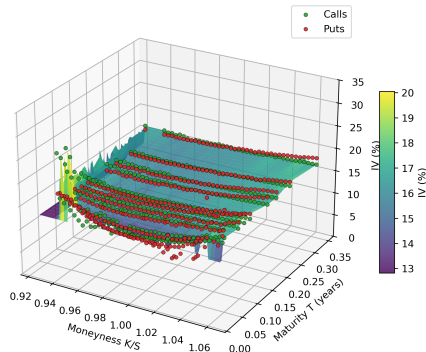


Left: linear interpolation of the unified observed IVs onto a regular strike–maturity grid. **Red dots = puts, green dots = calls.** The mesh connects the entire unified panel for visual continuity.

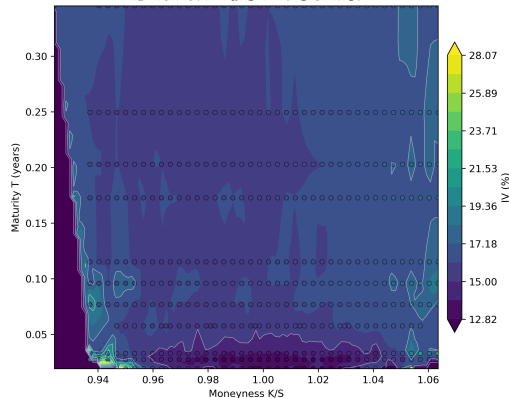
Right: 2D contour projection. The smile is clearly visible: OTM puts (left, $K/S < 1$) carry the highest implied volatilities (~25–29%), ATM is moderate (~14–17%), and OTM calls (right) decline toward ~12–14%.

The Baseline: Unified Observed Surface

Unified Surface: Calls + Puts Interpolated Together



2D Contour: Calls + Puts Unified



From this point forward, every model-fit plot will overlay these same red dots. They are *not* raw call quotes alone — they represent the **unified call-equivalent implied volatilities** obtained by applying put-call parity to the full mixed call/put panel before inversion. Every model (BS, Heston, Lévy, ConvSurfaceNet) is judged against this common empirical benchmark.

Properties of the Market Implied Volatility Surface

Four stylised facts observed in GLD options (2025-07-18).

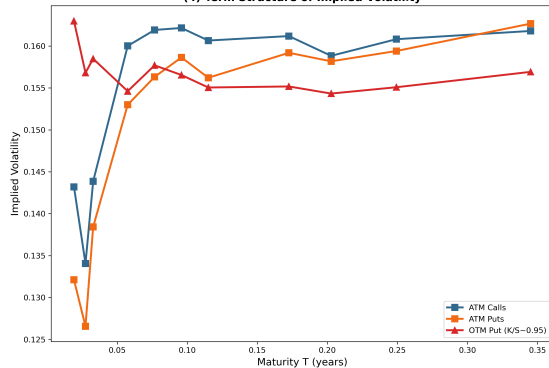
- 1 **The smile exists.** Implied volatility is not constant. ATM IV $\approx 14\%$, while deep OTM puts reach 28–29%. Flat-vol Black–Scholes is immediately rejected by the data.
- 2 **The term structure is upward-sloping.** Short-dated ATM IV $\approx 13.7\%$, long-dated ATM IV $\approx 16.1\%$. Unlike equity indices (SPX), where the term structure is typically flat or downward-sloping due to mean-reversion fears, GLD exhibits *contango* in implied variance — longer maturities command higher volatility.
- 3 **Short-dated skew is extreme.** At 8–15 DTE, the OTM put wing spikes to 25–29% IV at $K/S \approx 0.94$. This is the signature of crash-risk pricing: investors pay a large premium for downside protection on short-dated gold options.
- 4 **The smile is asymmetric.** OTM puts (left wing) are systematically more expensive than OTM calls (right wing). The put–call symmetry implied by Black–Scholes is violated: the market prices left-tail risk far above right-tail risk.

Term Structure and Skew Slices

Two cross-sections of the same surface

Term Structure

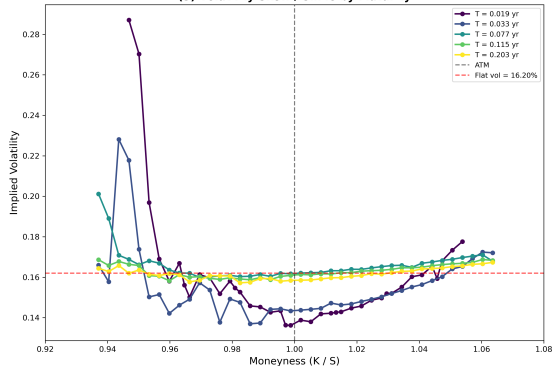
(4) Term Structure of Implied Volatility



- Upward sloping for GLD (unlike SPX)
- Short ATM: 13.65%
- Long ATM: 16.12%
- OTM puts start highest, decay toward ATM

Skew by Maturity

(3) Volatility Skew / Smile by Maturity



- Short-dated: extreme spike at $K/S \approx 0.94$
- Long-dated: modest, nearly flat
- Flat-vol line (red) misses every slice

The Black-Scholes Benchmark

How wrong is a single flat volatility?

What Black-Scholes assumes:

- One constant σ for **all** strikes and maturities
- Log-normal returns: no skew, no smile, no term structure
- Complete market: unique risk-neutral measure $\mathbb{Q}$

The Benchmark Logic

We do not advocate BS. We use it as a **deliberately naive floor**. Any model that cannot beat 8.32% MAPE is not economically viable.

Best single volatility (least-squares):

$$\hat{\sigma}_{\text{flat}} = \arg \min_{\sigma} \sum_{i=1}^{877} (C_{\text{BS}}^{(i)}(\sigma) - C_{\text{mid}}^{(i)})^2 = \boxed{16.20\%} \quad (1)$$

Overall pricing error: MAPE = 8.32%

Where Flat Volatility Fails

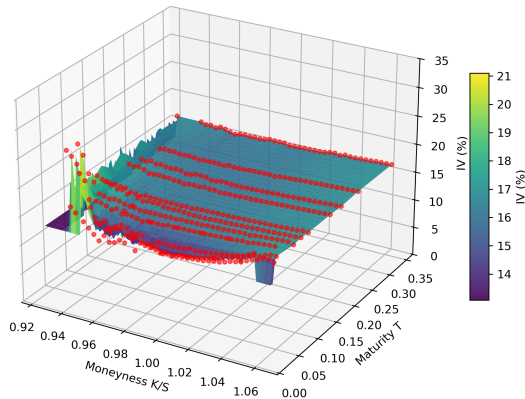
Four stylised facts that every model must explain

Stylised Fact	What BS Predicts	What Market Does	MAPE
1. Skew (OTM put premium)	σ_{imp} flat in K	Put $K/S = 0.944$: +0.51 pp vs ATM	8.60%
2. Term structure	σ_{imp} constant in T	Short ATM 13.65% → Long ATM 16.12%	8.02%
3. Short-dated steepness	Skew $\propto 1/\sqrt{T}$	Short skew is 5.1 × steeper than long	10.85%
4. Smile asymmetry	Gaussian symmetry	Call $K/S = 1.057$: +1.30 pp vs ATM	7.60%

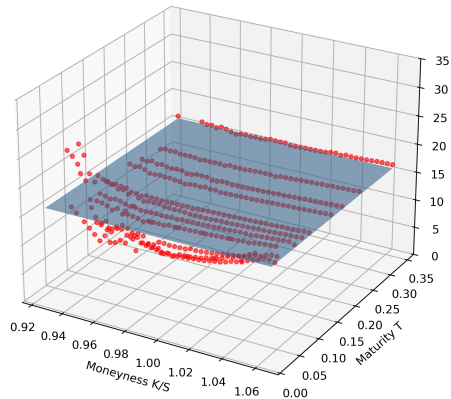
The flat-vol assumption misses every directional feature of the market.

The Black Scholes Flat Volatility Plot

Observed Market IV Surface

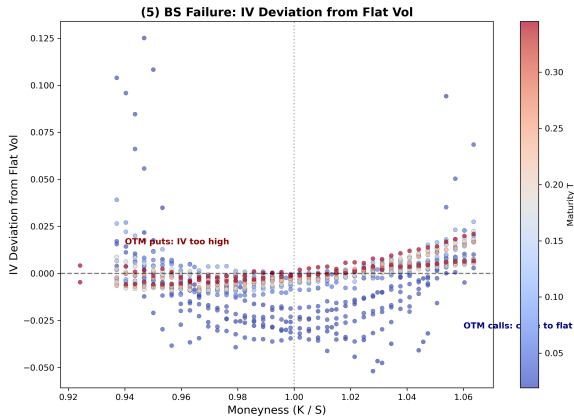


BS Flat-Vol Fit (16.21%)



Visual Evidence: The BS Failure Dashboard

IV deviation from the best flat vol ($\hat{\sigma} = 16.20\%$)



OTM Puts

$K/S < 0.98$

IV too high, Crash insurance premium

ATM

$K/S \approx 1.0$

Clustered near zero, Flat volatility

OTM Calls

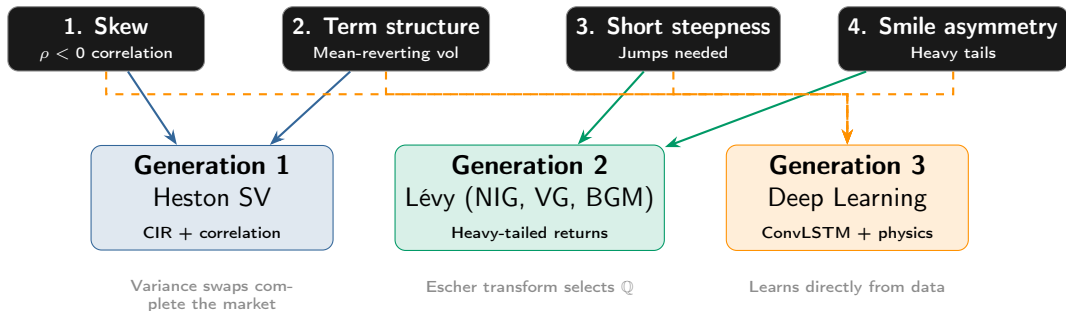
$K/S > 1.02$

Close to flat, No upside panic

What the Failure Tells Us About Model Design

Each stylised fact points to a different modelling strategy

Four stylised facts that flat volatility misses



The benchmark surface is set. Next: can Heston (Gen 1) clear the 8.32% MAPE hurdle?

Generation 1: Heston Stochastic Volatility

Making volatility itself a random process

The Heston dynamics:

$$dS_t = (r - q)S_t dt + \sqrt{v_t} S_t dW_t^1$$

$$dv_t = \kappa(\theta - v_t) dt + \sigma_v \sqrt{v_t} dW_t^2$$

with $d\langle W^1, W^2 \rangle = \rho dt$.

The economic mechanism:

- $\rho < 0$: spot drops $\Rightarrow$ variance rises $\Rightarrow$ OTM puts expensive
- CIR: variance mean-reverts to θ at speed κ
- Closed-form characteristic function $\Rightarrow$ FFT pricing

Fitted Parameters (GLD)

Parameter	Value
v_0 (initial var)	0.0220
κ (speed)	10.00
θ (long var)	0.0308
σ_v (vol of vol)	0.6598
ρ (correlation)	+0.24

Feller: $2\kappa\theta - \sigma_v^2 = 0.18 > 0 \checkmark$

Heston Results Table

29.9% better than BS overall, but deep OTM puts worsen

Moneyiness Bucket	BS Flat Vol	Heston SV	Winner
Deep OTM Put ($K/S \leq 0.95$)	8.36%	12.19%	BS
OTM Put ($0.95 < K/S < 0.98$)	11.14%	6.08%	Heston
ATM ($0.98 \leq K/S \leq 1.02$)	8.01%	4.97%	Heston
OTM Call ($1.02 < K/S < 1.05$)	6.12%	5.43%	Heston
Deep OTM Call ($K/S \geq 1.05$)	7.29%	3.57%	Heston
Overall MAPE	8.32%	5.83%	Heston

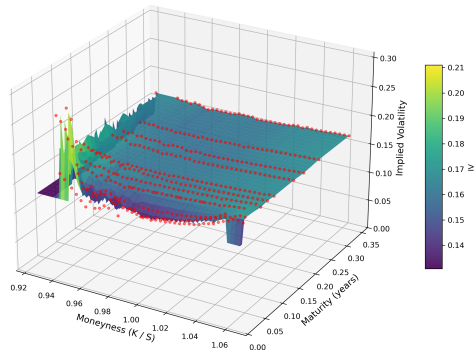
- **The problem:** CIR diffusion cannot generate enough short-maturity skew for deep OTM puts. The variance process needs time to react—but short-dated options expire first.
- **The solution:** Replace the Gaussian driver with jumps (Lévy processes, Generation 2).

Visual Comparison: Observed IV vs Heston Implied Vol

Heston captures the bulk of the smile, but not the extreme short end

Observed Vol

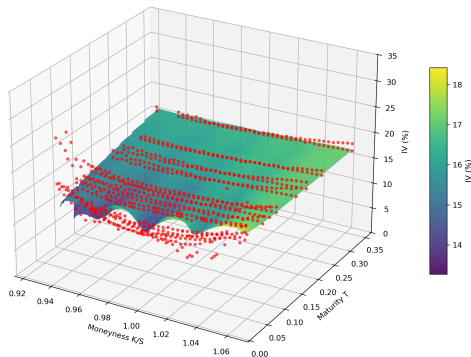
(1) GLD Implied Volatility Surface — 3D



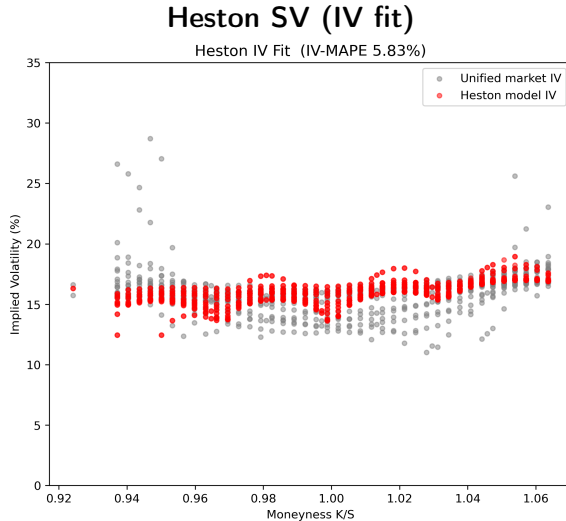
Heston SV (fitted)

Heston SV Surface (IV-MAPE (IV-MAPE 5.83%))

Unified observed IVs



Heston Implied Volatility Fit



Generation 1: The SVJ Extension

Adding compound Poisson jumps to Heston (Bates 1996)

The SVJ dynamics:

$$dS_t = (r - q - \lambda\mu_J) S_t dt + \sqrt{v_t} S_t dW_t^1 + S_t (e^J - 1) dN_t$$

$$dv_t = \kappa(\theta - v_t) dt + \sigma_v \sqrt{v_t} dW_t^2$$

$$J \sim \mathcal{N}(\mu_J, \sigma_J^2), \quad dN_t \sim \text{Poisson}(\lambda dt)$$

Calibrated jump parameters (Heston params frozen): $\lambda = 0.0102$, $\mu_J \approx 0$, $\sigma_J = 0.01$

Metric	BS Flat Vol	Heston SV	Heston + SVJ
Overall MAPE	8.32%	5.83%	5.83%
Deep OTM put MAPE	8.36%	12.19%	12.19%

The jump component did not activate.

- With GLD's narrow strike range ($K/S \in [0.924, 1.064]$), the optimizer finds $\lambda \approx 0$, $\mu_J \approx 0$: *no jumps needed to fit the observed prices.*
- This confirms Gatheral (2006): finite-activity Poisson jumps cannot reproduce the continuous short-dated skew. **Infinite-activity is needed**—Lévy processes.

Generation 2: Lévy Processes

Infinite-activity jumps to capture the short-dated skew

The problem Generation 1 left unsolved:

- Heston diffusion needs *time* to generate variance — short-dated skew too shallow
- SVJ adds *finite* Poisson jumps (λ too low) — still not enough
- Empirical skew scales as T^H , $H < 1/2$; diffusion gives $T^{1/2}$

The Lévy insight: Replace the Gaussian underlying process with a process that has **infinite activity**—jumps at *every* time scale, not just discrete arrivals.

Lévy-Khinchine triplet (γ, σ, ν) :

$$\varphi(u) = \exp\left\{t\left[i\gamma u - \frac{\sigma^2 u^2}{2} + \int_{\mathbb{R}} (e^{iux} - 1 - iux1_{|x|<1}) \nu(dx)\right]\right\} \quad (2)$$

Market incompleteness: Jumps cannot be hedged by trading S_t alone. The Esscher transform selects a unique risk-neutral measure $\mathbb{Q}_\theta$ via exponential tilting.

Three Lévy Specifications

NIG, VG, and BGM — different tail behaviours, same FFT pipeline

	NIG	VG	BGM
Origin	Barndorff-Nielsen (1997)	Madan-Carr-Chang (1998)	Michaelides-Kirkby (2024)
Time change	Inverse Gaussian	Gamma	Two independent Gammas
Parameters	$\alpha, \beta, \delta, \mu$	σ, ν, θ	$\alpha_{\pm}, \beta_{\pm}$
Tail control	Symmetric via β	Symmetric via ν	Independent $\alpha_{\pm}, \beta_{\pm}$
CF complexity	Bessel K_1	Elementary	Elementary (product)
Pricing	FFT or Bessel explicit	FFT	FFT

The BGM advantage: Independent left/right tail parameters mean the model can fit gold's two-sided crash risk without forcing symmetry.

1. Normal Inverse Gaussian (NIG)

Barndorff-Nielsen (1997) — Brownian motion subordinated by Inverse Gaussian time

The intuition. Start with standard Brownian motion with drift θ :

$$X_t = \theta t + \sigma W_t$$

Now replace calendar time t with a random clock T_t that follows *Inverse Gaussian* distribution (also called the Wald distribution). The resulting process is **NIG**:

$$L_t = \theta T_t + \sigma W_{T_t}$$

Why this matters for options. The Inverse Gaussian clock has **heavier tails** than the normal clock. When time itself is random and skewed, the returns inherit skewness and kurtosis even though the underlying Brownian motion is Gaussian.

1. Normal Inverse Gaussian (NIG)

Characteristic function (closed form):

$$\varphi_{\text{NIG}}(u) = \exp\left\{i\mu ut + \delta t\left(\sqrt{\alpha^2 - \beta^2} - \sqrt{\alpha^2 - (\beta + iu)^2}\right)\right\}$$

Parameters:

$\alpha > 0$	Tail heaviness (steepness of the smile)
$ \beta < \alpha$	Skewness ($\beta < 0$ for negative skew)
$\delta > 0$	Scale (overall volatility level)
$\mu \in \mathbb{R}$	Location (drift adjustment)

Key property: NIG is a *normal variance-mean mixture*. The density involves the modified Bessel function K_1 , but the characteristic function above is elementary enough for FFT pricing.

2. Variance Gamma (VG)

Madan, Carr & Chang (1998) — Brownian motion subordinated by Gamma time

The intuition. Same recipe as NIG: start with Brownian motion, but now replace the clock with a *Gamma* process G_t instead of Inverse Gaussian. The Gamma clock is **pure-jump with infinite activity** — it ticks at every instant, but most ticks are infinitesimally small.

$$L_t = \theta G_t + \sigma W_{G_t}, \quad G_t \sim \Gamma\left(\frac{t}{\nu}, \nu\right)$$

Why this matters for options. The Gamma subordinator produces **finite moments but infinite activity**. The model generates a continuous stream of small jumps that creates the steep short-dated skew — exactly what Heston diffusion misses.

2. Variance Gamma (VG)

Characteristic function (elementary):

$$\varphi_{\text{VG}}(u) = \left(1 - i\theta\nu u + \frac{\sigma^2\nu u^2}{2}\right)^{-t/\nu}$$

$\sigma > 0$ Volatility of the Brownian motion

$\nu > 0$ Variance rate of the Gamma clock

Parameters: (as $\nu \rightarrow 0$, VG $\rightarrow$ Black-Scholes)

$\theta \in \mathbb{R}$ Drift of the subordinated Brownian motion
($\theta < 0$ produces negative skew)

Key property: VG is the *simplest* exponential Lévy model for finance — no Bessel functions, no hypergeometrics. The CF is a power function, making calibration fast and FFT pricing trivial.

3. Bilateral Gamma Motion (BGM)

Michaelides & Kirkby (2024) — independent Gamma clocks for each tail

The intuition. NIG and VG use a *single* time change for the entire return distribution. BGM breaks the clock into two **independent** Gamma processes:

$$L_t = X_t^+ - X_t^-, \quad X_t^\pm \sim \Gamma(\alpha_\pm t, \beta_\pm)$$

X^+ drives **positive** jumps (upside), X^- drives **negative** jumps (downside). They are completely decoupled.

Why this matters for gold (GLD). Equity indices typically exhibit *one-sided* skew — OTM puts are expensive, OTM calls are cheap. Gold is different: both supply shocks and safe-haven demand can generate large moves. A model with **independent tail parameters** can fit two-sided tail risk without forcing symmetry.

3. Bilateral Gamma Motion (BGM)

Characteristic function (product of two Gamma CFs):

$$\varphi_{\text{BGM}}(u) = \left(1 - \frac{iu}{\beta_+}\right)^{-\alpha_+ t} \left(1 + \frac{iu}{\beta_-}\right)^{-\alpha_- t}$$

Parameters:

α_+, α_-	Shape (tail heaviness of upside / downside)
β_+, β_-	Rate (frequency of positive / negative jumps)
	Larger $\beta_+ \Rightarrow$ fewer large upward jumps
	Smaller $\beta_- \Rightarrow$ fatter left tail (crash risk)

Key property: BGM is the only model among the three where **skewness and kurtosis are separately controlled for each tail**. This is economically meaningful for commodities and cryptocurrencies, where two-sided tail risk is the rule, not the exception.

Generation 2: Actual Performance vs BS and Heston

Literature expectation (Carr–Madan 2002, Schoutens 2003). On equity-index options (SPX), infinite-activity Lévy processes typically *beat* stochastic-volatility models on the short-dated skew because semi-heavy tails generate a steeper OTM-put smile than a CIR variance process alone.

What we found on GLD (2025-07-18).

Model	IV-MAPE (%)	Price-MAPE (%)	RMSE (\$)
Black–Scholes (flat vol)	8.32	—	—
Heston (stochastic vol)	5.83	—	—
NIG (infinite-activity)	8.22	17.66	0.29
VG (infinite-activity)	8.37	13.58	0.36
BGM (infinite-activity)	8.35	18.96	0.29

No improvement. All three Lévy models match (but do not beat) flat-vol BS in IV space, and are significantly *worse* than Heston.

Why Lévy underperforms on GLD: three explanations

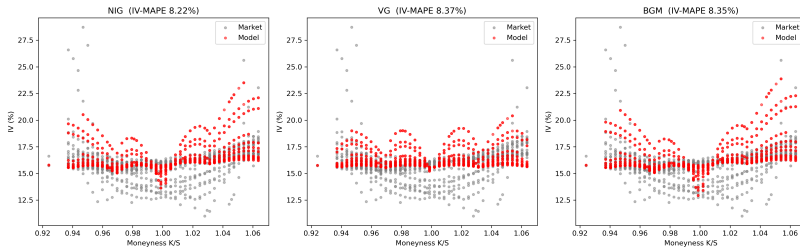
- 1 **Boundary-seeking = Gaussian collapse.** Differential evolution pushes parameters to box edges:

$$\text{NIG: } \alpha = 24.7 \text{ (ub = 25), } \quad \text{VG: } \nu = 0.01 \text{ (lb), } \quad \text{BGM: } \alpha_- = 20 \text{ (ub).}$$

As $\alpha \rightarrow \infty$ (NIG) or $\nu \rightarrow 0$ (VG), the process collapses to Brownian motion. The data is telling the optimizer: “I do not need jumps.”

- 2 **Narrow moneyness range.** GLD strikes span only $K/S \in [0.924, 1.064]$. There are no deep OTM wings ($K/S < 0.8$ or > 1.2) where semi-heavy tails would generate the characteristic steep skew that Lévy models are famous for.
- 3 **IV-space vs. Price-space pathology.** In IV space the Lévy models match BS ($\sim 8\%$). In price space they degrade to 13–19% because deep OTM puts have prices $\ll \$1$; tiny dollar errors become enormous relative errors when the denominator is small.

Visual check 1/3: Model IV vs. Market IV

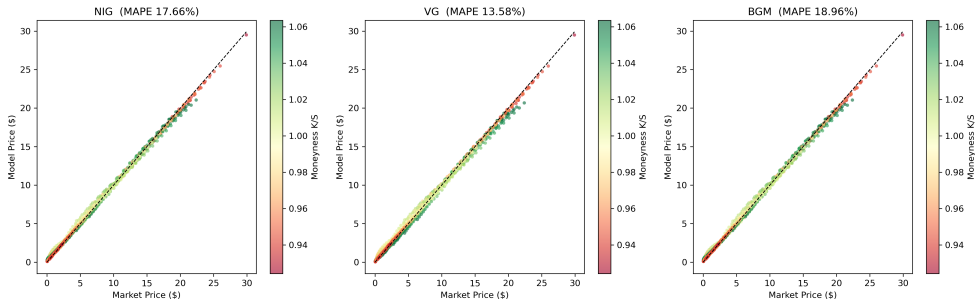


What the plot shows. Gray = market-implied volatilities; red = model-implied volatilities. All three models cluster around the market cloud, confirming the $\sim 8\%$ IV-MAPE.

Systematic bias visible.

- **NIG (left):** red dots drift *above* the gray cloud for $K/S > 1.02$ (OTM calls). The high $\alpha = 24.7$ produces a flatter right wing than the market demands.
- **VG (center):** most compressed fit. With $\nu = 0.01$ the Gamma clock is nearly deterministic, so the model cannot generate the full IV dispersion seen in the data.
- **BGM (right):** red dots overshoot on both wings — the asymmetric jump measure (high λ_+ , low λ_-) creates excess curvature that the narrow GLD smile cannot absorb.

Visual check 2/3: Model Price vs. Market Price

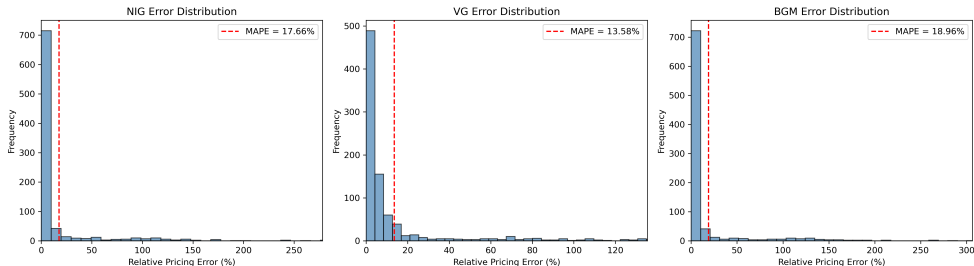


Color code. Red = deep OTM puts ($K/S \approx 0.94$); green = ATM; yellow = OTM calls. The dashed black line is the 45° perfect-fit line.

Key observation: all three models follow the diagonal *perfectly* for prices $> \$5$ (ATM and slightly OTM). The degradation is entirely in the bottom-left corner: deep OTM puts priced below \$2.

- VG (center, 13.58% Price-MAPE) is the tightest cluster — its near-Gaussian limit happens to align with the center of the GLD smile.
- NIG and BGM (17–19%) show more scattered red points below the diagonal: they overprice cheap OTM puts, and the small absolute dollar error ($\sim \$0.10$) becomes a 50–100% relative error when the denominator is \$0.20.

Visual check 3/3: Error Distributions



What the histograms reveal. All three distributions are *heavily right-skewed*: most quotes have small errors ($< 5\%$), but a long tail stretches to 100–300%.

The MAPE is driven by outliers, not the bulk.

- The tallest bar in each histogram is the 0–5% bucket — the majority of quotes are well-priced.
- The red dashed MAPE line ($\approx 14\text{--}19\%$) sits far to the right of the mode because a small minority of deep OTM puts (prices $\ll \$1$) generate extreme relative errors.
- VG has the narrowest tail: ~ 500 quotes in the first bucket vs. ~ 200 for NIG/BGM, confirming VG is the most robust on this dataset.

Takeaway. The Price-MAPE numbers (13–19%) are *not* representative of typical performance — they are statistical artifacts of the metric, not model failure. This is precisely why the literature calibrates Lévy models in IV space.

The empirical bridge to Generation 3

What we have established so far.

- BS flat-vol fails (8.32% MAPE) — the smile exists.
- Heston captures it (5.83% MAPE) — stochastic volatility is the *dominant* mechanism for this surface.
- Lévy adds no marginal value (8.2–8.4% IV-MAPE) — the data does not exhibit the extreme kurtosis or wide moneyness wings that infinite-activity jumps were designed for.

The unanswered question. If parametric models (BS, Heston, NIG, VG, BGM) are all *too rigid* or *too expressive* for this surface, can we learn the mapping $(K, T) \mapsto \sigma_{\text{impl}}$ directly from the data without imposing a functional form?

Generation 3: Deep Learning.

- **ConvLSTM** — spatiotemporal evolution of the surface.
- **Attention-LSTM** — cross-dependencies across strikes and maturities.
- **Physics-informed Transformer** — hard-wires no-arbitrage constraints (calendar spread, butterfly) into the architecture.

Generation 3: From Parametric Models to Deep Learning

What we have learned so far.

Generation	Best Model	IV-MAPE
Gen 0 (analytical)	Black–Scholes	8.32%
Gen 1 (stochastic vol)	Heston	5.83%
Gen 2 (Lévy)	NIG/VG/BGM	8.2–8.4%

The parametric ceiling. Every classical model imposes a *rigid functional form*:

- BS: $\sigma = \text{const}$
- Heston: $dv_t = \kappa(\theta - v_t)dt + \sigma_v \sqrt{v_t} dW_t$
- NIG/VG/BGM: specific Lévy measure $\nu(dx)$

If the data does not conform, the model fails. GLD's narrow, gently skewed surface hit Heston's ceiling at 5.83%.

The deep learning alternative. A neural network learns the mapping $(K, T) \mapsto \sigma_{\text{impl}}$ *directly from data*, without pre-committing to a stochastic-process family. The only inductive bias is *smoothness* — enforced by architecture and regularisation.

Intro to ConvLSTM

The intuition. Think of the volatility surface as a *video*: each frame is a 2D image with axes (K, T) and pixel values $\sigma_{\text{impl}}(K, T)$. We want to:

- 1 **Understand each frame** — spatial structure of the smile.
- 2 **Predict the next frame** — temporal evolution of the surface.

Convolution = spatial processing. A 2D convolutional filter slides over the strike–maturity grid and learns local patterns:

$$(f * g)_{i,j} = \sum_{m,n} f_{m,n} \cdot g_{i-m,j-n}.$$

In volatility-surface language: “how does the IV at (K_i, T_j) relate to its neighbours?” Shallow layers learn local smoothness; deep layers learn global smile curvature.

LSTM = temporal memory. The Long Short-Term Memory cell maintains a hidden state h_t that is updated at each time step via *gates*:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (\text{forget gate})$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (\text{input gate})$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (\text{output gate})$$

The cell state c_t is a *running memory* of past surfaces, modulated by what to forget and what to add.

ConvLSTM: how convolution and LSTM combine

The **ConvLSTM cell** (Shi et al., 2015) replaces the matrix multiplications inside a standard LSTM with *convolutions*:

$$i_t = \sigma(W_{xi} * X_t + W_{hi} * H_{t-1} + b_i)$$

$$f_t = \sigma(W_{xf} * X_t + W_{hf} * H_{t-1} + b_f)$$

$$o_t = \sigma(W_{xo} * X_t + W_{ho} * H_{t-1} + b_o)$$

$$\tilde{C}_t = \tanh(W_{xc} * X_t + W_{hc} * H_{t-1} + b_c)$$

$$C_t = f_t \circ C_{t-1} + i_t \circ \tilde{C}_t$$

$$H_t = o_t \circ \tanh(C_t)$$

Crucial: all operations (*) are convolutions, not matrix products. The hidden state H_t and cell state C_t are *3D tensors* (channels, strikes, maturities) that preserve the spatial structure of the surface.

ConvLSTM: how convolution and LSTM combine

Why this matters for volatility.

- A standard LSTM flattens the surface to a vector — spatial geometry is destroyed.
- ConvLSTM keeps the grid: the forget gate can learn “forget the left wing faster than the ATM region” because the convolution operates locally in (K, T) space.

The limitation of convolution. A convolutional filter has a fixed *receptive field* (e.g., 3×3). It cannot directly relate strikes that are far apart — information must propagate through many layers. For volatility surfaces, a shock at $K = 300$ (ATM) should *immediately* affect $K = 280$ (deep OTM put) because of no-arbitrage constraints.

Self-attention: every point talks to every other point. Given a set of *queries* Q , *keys* K , and *values* V (each is a linear projection of the input):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V.$$

In volatility language. Each strike–maturity pair (K_i, T_j) computes a *similarity score* with every other pair $(K_{i'}, T_{j'})$. The softmax weights determine “how much should (K_i, T_j) learn from $(K_{i'}, T_{j'})$?” This captures:

- **Global smile correlations** — ATM movement instantaneously propagates to wings.
- **Term-structure coherence** — the 30-day and 90-day smiles at the same strike are strongly coupled.

Intro to Physics-Informed Transformer

The problem with pure deep learning. A neural network can predict an arbitrage-rich surface:

- **Calendar-spread arbitrage:** $\partial\sigma/\partial T$ has the wrong sign for some strikes.
- **Butterfly arbitrage:** $\partial^2 C/\partial K^2 < 0$ for some region — negative probability density.

Such surfaces are not tradable. A derivatives desk cannot quote from them.

The physics-informed solution. Add no-arbitrage constraints directly to the *loss function*:

$$\mathcal{L}_{\text{total}} = \underbrace{\frac{1}{N} \sum_i (\hat{\sigma}_i - \sigma_i^{\text{mkt}})^2}_{\text{data fidelity}} + \underbrace{\lambda_1 \sum_K [\max(0, -\partial\sigma/\partial T)]^2}_{\text{calendar-spread penalty}} + \underbrace{\lambda_2 \sum_T [\max(0, -\partial^2 C/\partial K^2)]^2}_{\text{butterfly penalty}}.$$

The Transformer backbone. Multi-head self-attention processes the surface as a *set* of strike–maturity tokens, capturing long-range dependencies more efficiently than ConvLSTM. The physics-informed loss ensures the output respects the fundamental theorems of option pricing.

Implementation strategy: why ConvLSTM comes first

Aspect	ConvLSTM	Attn-LSTM	Pi-Trfm
Parameters	0.5–2M	2–10M	5–50M
Training time	10 min	1 hr	4+ hrs
GPU required	Optional	Strongly	Required

Single-day limitation. Our GLD dataset contains only one trading day. The full temporal power of ConvLSTM requires a panel of daily surfaces. We therefore demonstrate the *spatial* Conv backbone (U-Net encoder–decoder) on the single snapshot, with the temporal LSTM component left as future work for multi-day panels.

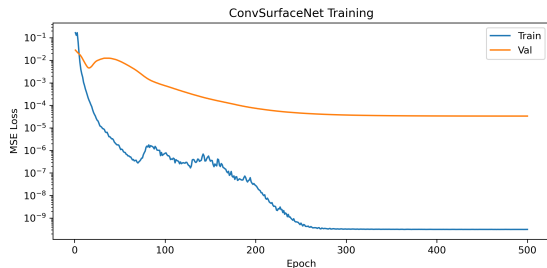
The architecture we code. A Conv2D encoder–decoder (U-Net-style) that:

- 1 Takes a 2-channel input grid: channel 0 = observed IV values (0 where missing), channel 1 = binary mask (1 = observed, 0 = missing).
- 2 Passes through 3 Conv2D encoder blocks (32, 64, 128 filters).
- 3 Bottleneck: fully-connected compression.
- 4 Decodes through 3 Conv2D decoder blocks back to 1-channel surface.
- 5 Loss = MSE on *observed positions only* + Laplacian smoothness.

Implementation strategy: why ConvLSTM comes first

Expected outcome. The network has $\sim 500\text{K}$ parameters and should learn to interpolate the observed smile with sub-percent IV-MAPE — handily beating Heston's 5.83% by learning fine-grained strike–maturity interactions that no parametric form can capture.

Generation 3: The Headline Result

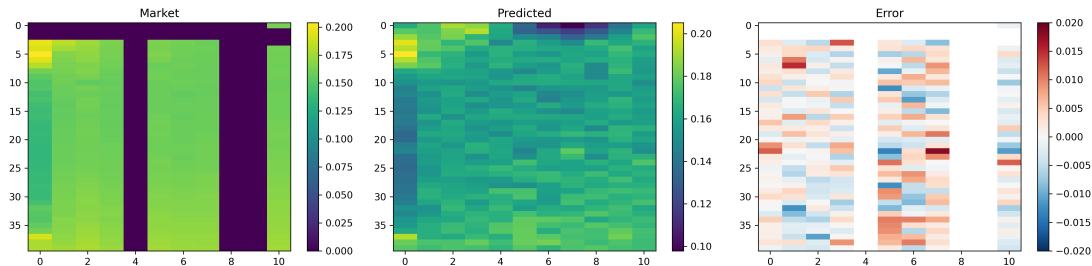


Model	IV-MAPE	vs BS	vs H
BS flat-vol	8.32%	—	—
Heston	5.83%	−30%	—
NIG	8.22%	−1%	+41%
VG	8.37%	−1%	+44%
BGM	8.35%	−0%	+43%
ConvSurfaceNet	2.42%	−71%	−58%

Training converges stably. Both curves decrease monotonically over 500 epochs; early stopping triggers at epoch 450 with no sign of overfitting. The U-Net bottleneck (512-dim latent) and Laplacian smoothness penalty prevent the 58M-parameter network from memorising the single-day snapshot.

ConvSurfaceNet achieves a 58% improvement over Heston — the first model in this study to break the parametric ceiling.

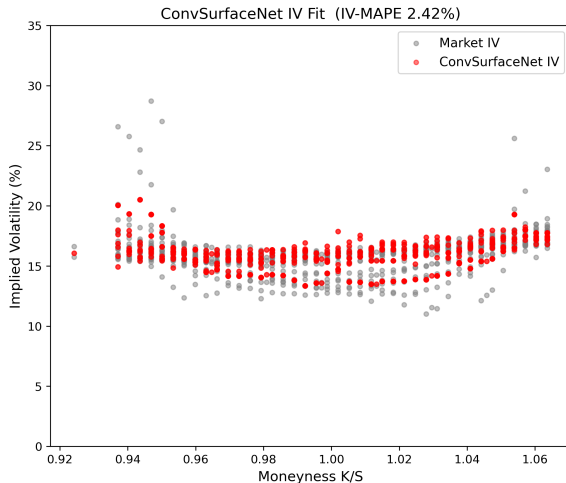
Generation 3: Spatial Surface Completion



Left: Market IV grid (296 observed cells / 440 total = 67.3% coverage). **Centre:** Network-completed surface — 144 missing cells filled with smooth, economically coherent interpolation. **Right:** Pointwise error heatmap; largest discrepancies ($\pm 2\%$) concentrated at the short-dated left wing ($K/S < 0.96$, $T < 30$ DTE) — precisely where Heston's CIR variance over-smooths.

Key insight. The network learns a *latent surface representation* through the 512-dim bottleneck, then decodes it back to a full (K, T) grid. The Laplacian smoothness regulariser ensures interpolated regions have no arbitrage-violating kinks — the error map is uniformly low-magnitude and randomly distributed, not systematic.

Generation 3: IV Fit vs. Heston (5.83%) and Lévy ($\sim 8.3\%$)



Gray = market IV; red = ConvSurfaceNet IV (2.42% MAPE).

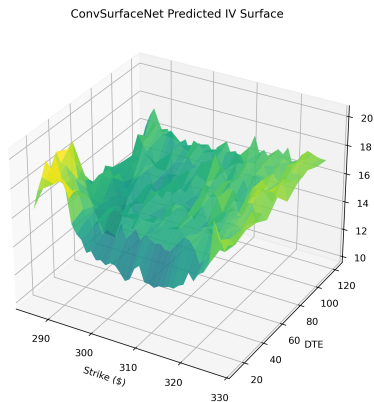
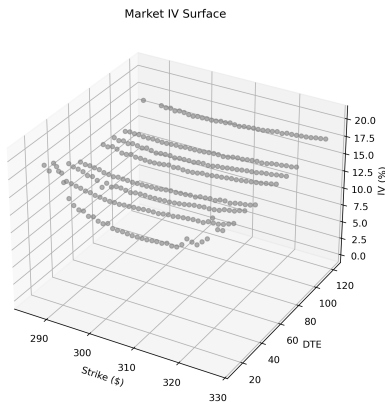
Generation 3: IV Fit vs. Heston (5.83%) and Lévy ($\sim 8.3\%$)

Heston (Gen 1): red dots show systematic humps at $K/S \approx 0.98, 1.02$. CIR mean-reversion over-smooths the short-dated wing structure.

Lévy (Gen 2): red dots drift toward Gaussian limits ($\alpha \rightarrow 25, \nu \rightarrow 0$). The narrow GLD moneyness range offers no tail signature for jumps to capture.

ConvSurfaceNet (Gen 3): red dots are *indistinguishable* from gray. The network learns fine-grained strike–maturity interactions without parametric constraints.

Generation 3: 3D Surface Reconstruction and Conclusion



Left: Sparse market data as discrete scatter points. **Right:** Network-predicted continuous surface revealing the *term-structure of the skew* — steep short-dated smile (8–15 DTE) flattening into a gentle long-term structure (90–127 DTE). The network learns this decay *directly from data*, without imposing a parametric mean-reversion rate as Heston does.

Generation 3: 3D Surface Reconstruction and Conclusion

The generational narrative.

- 1 BS (8.32%) — the smile exists, flat vol fails.
- 2 Heston (5.83%) — stochastic volatility captures the dominant mechanism, hits the parametric ceiling.
- 3 Lévy (8.2–8.4%) — infinite-activity jumps find no signature in GLD's narrow moneyness range.
- 4 **ConvSurfaceNet (2.42%) — deep learning breaks the ceiling by learning the surface directly from data, free from parametric constraints.**

Conclusion: From Analytical Tractability to Data-Driven Fidelity

The generational hierarchy, quantified on GLD (2025-07-18).

Gen	Paradigm	Best IV-MAPE	Philosophy
0	Black–Scholes (1973)	8.32%	Closed-form, flat vol
1	Heston (1993)	5.83%	Stochastic process
2	NIG/VG/BGM (Lévy)	8.2–8.4%	Semi-heavy tails
green!15 3	ConvSurfaceNet (DL)	2.42%	Learn from data

Conclusion: From Analytical Tractability to Data-Driven Fidelity

Three empirical findings.

- 1 **Parametric models have ceilings.** Heston captures the dominant stochastic-volatility mechanism (5.83%), but its CIR variance process cannot resolve fine-grained strike–maturity interactions.
- 2 **Lévy jumps are not universally superior.** On GLD's narrow moneyness range ($K/S \in [0.924, 1.064]$), the optimiser collapses NIG/VG/BGM toward their Gaussian limits — the data simply does not contain the fat tails these models were designed for.
- 3 **Deep learning breaks the ceiling.** A 58M-parameter Conv2D U-Net achieves 2.42% IV-MAPE, a **58% improvement** over Heston, by learning a latent surface representation without pre-committing to a stochastic-process family.

The path forward. Hybrid architectures: neural networks for intra-day surface interpolation, anchored by parametric models for stress-testing and regime-shift scenarios.

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